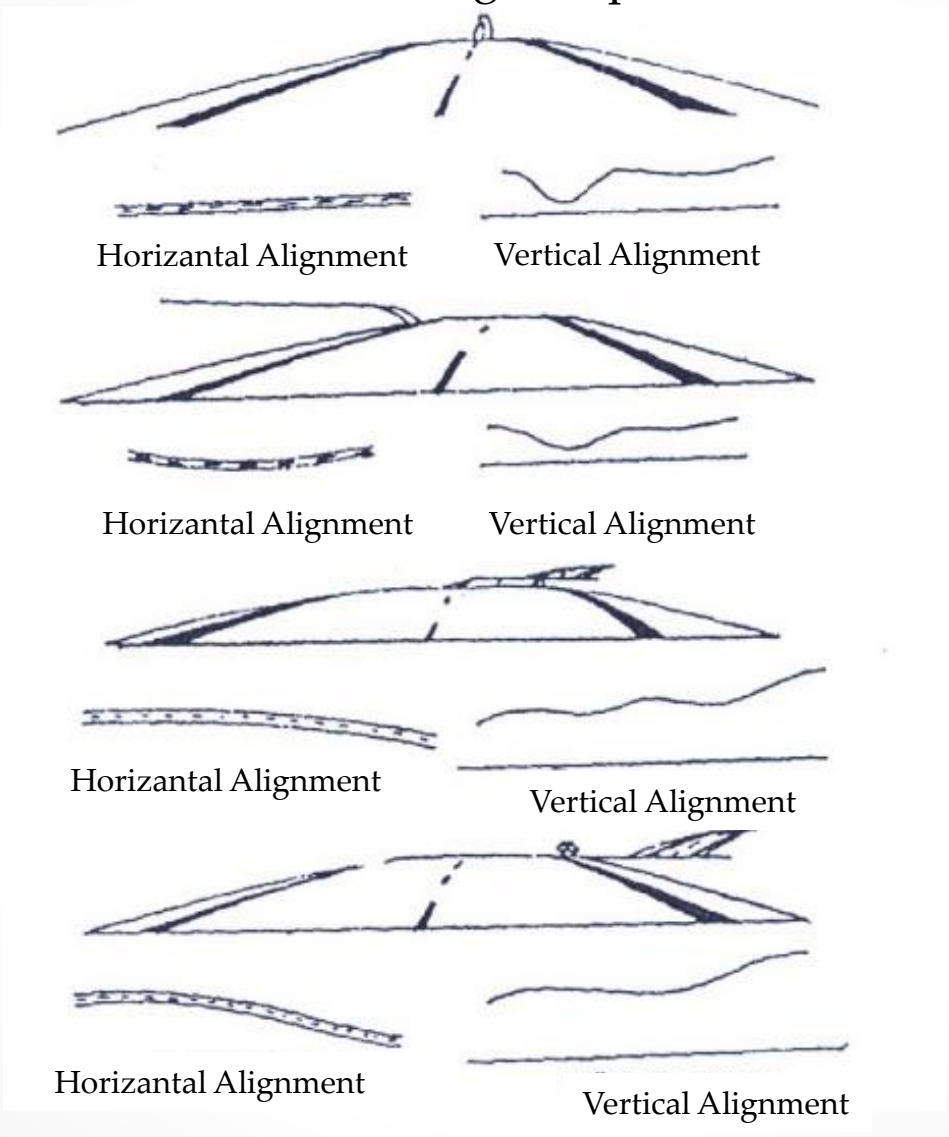
# INS3441

# HIGHWAY ENGINEERING

**Prof. Halit ÖZEN, PhD.**

# Alignment and Profile Relationships in Roadway Design

Hidden – Downgrade profile



# Alignment and Profile Relationships in Roadway Design

Short humps in the grade should be avoided





Horizantal Alignment



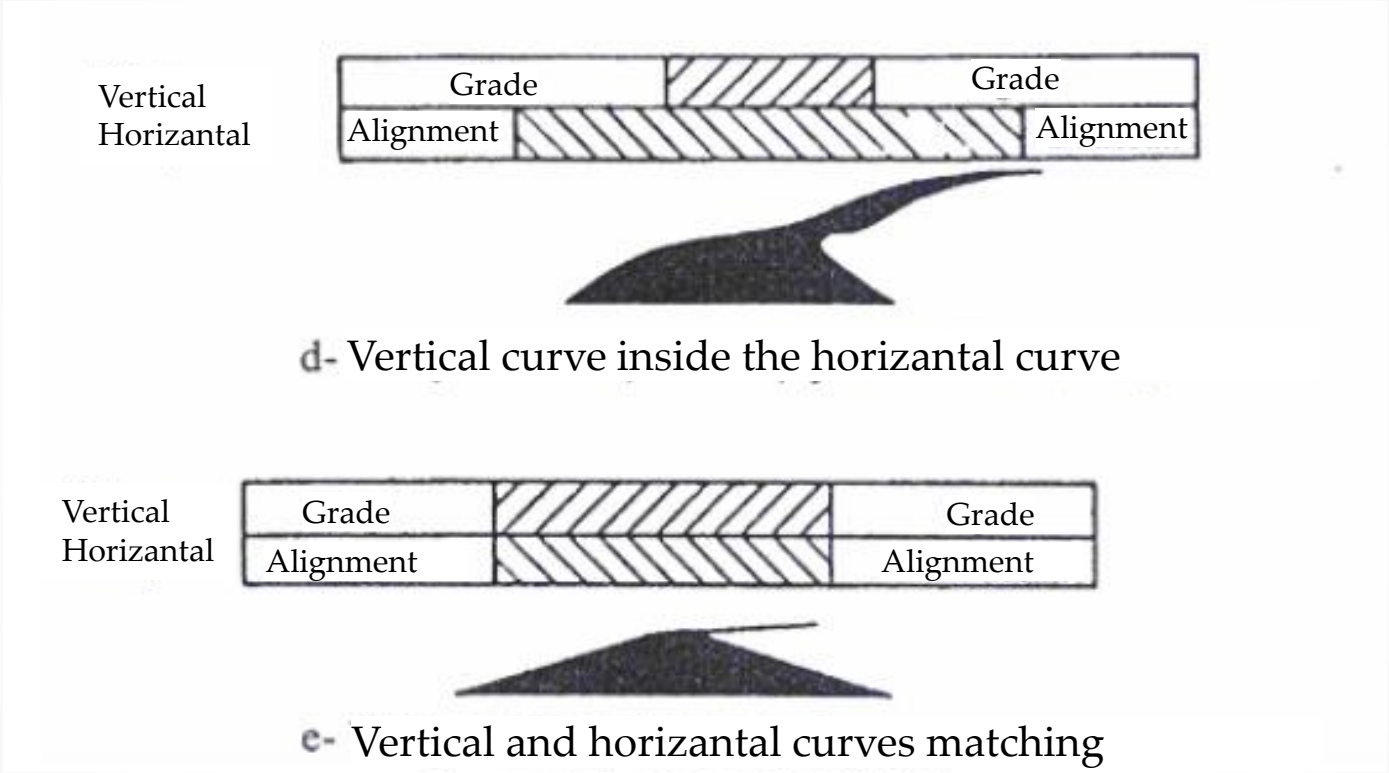
Vertical Alignment

Short alignment between same directional alignment



# Alignment and Profile Relationships in Roadway Design

## Coordination of Horizantal and Vertical Curve



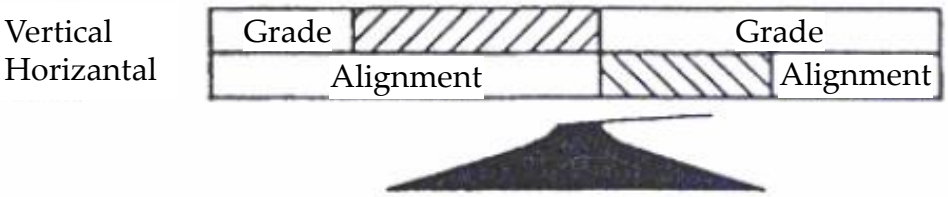

d- Vertical curve inside the horizantal curve



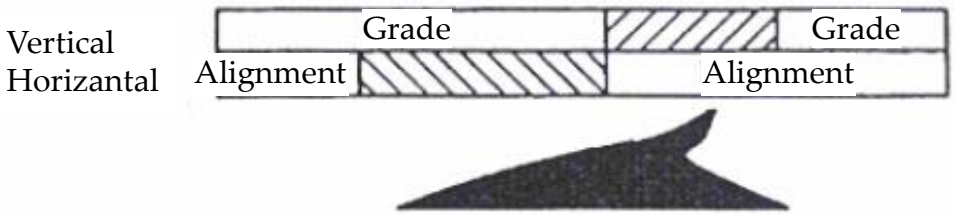

e- Vertical and horizantal curves matching

# Alignment and Profile Relationships in Roadway Design

## Coordination of Horizantal and Vertical Curve




a- Vertical curve at the begining of horizontal curve



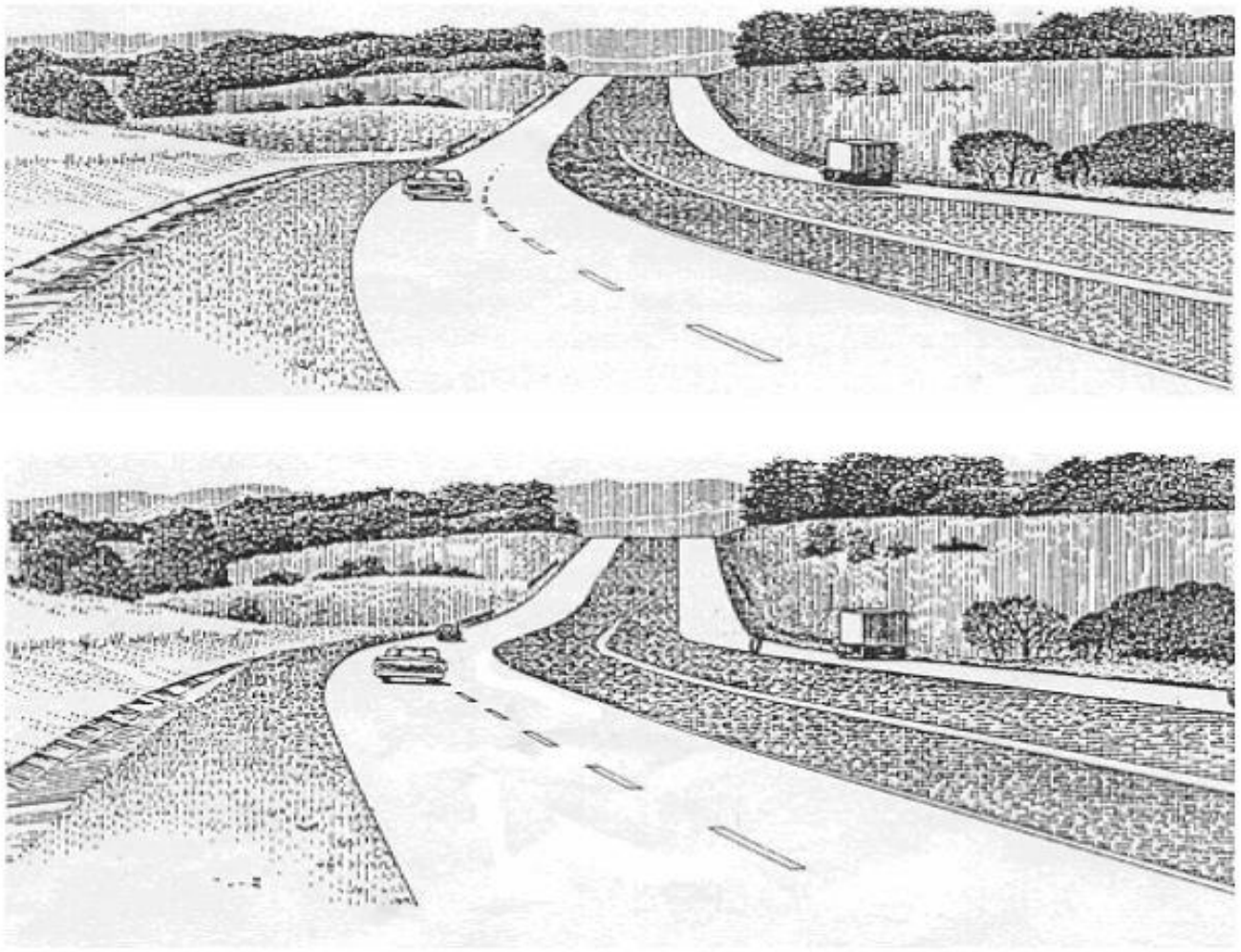

b- Vertical curve at the end of horizontal curve




c- Vertical and horizantal curve match at the begining of horizontal curve

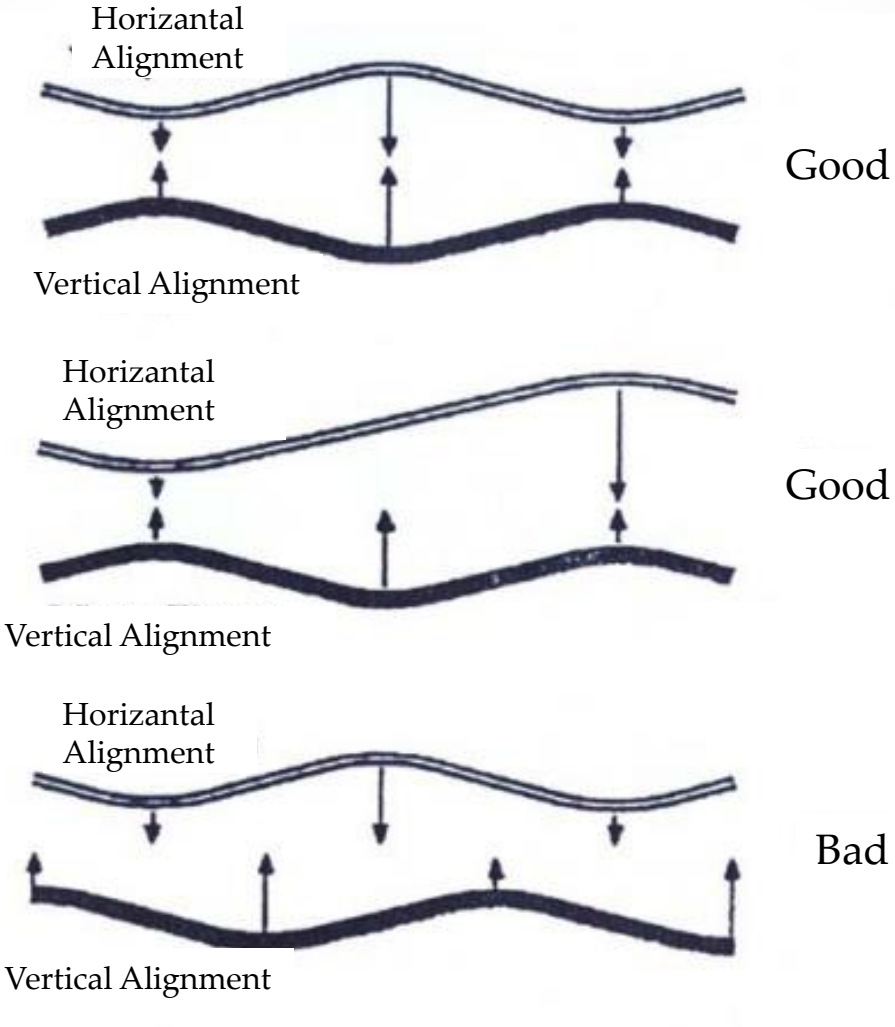
# Alignment and Profile Relationships in Roadway Design

Short vertical curve after long vertical curve



# Alignment and Profile Relationships in Roadway Design

## Horizontal/Vertical Curve Relationships

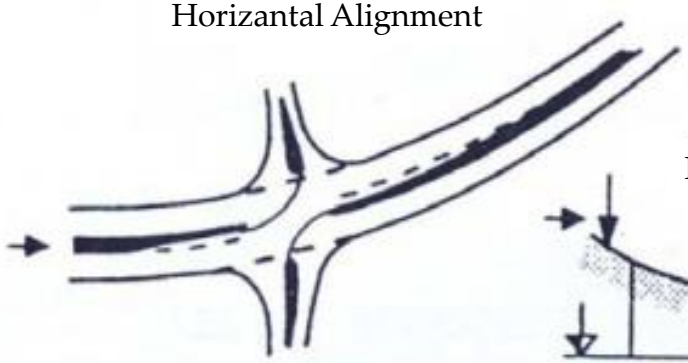


# Alignment and Profile Relationships in Roadway Design

Visibility of Intersection

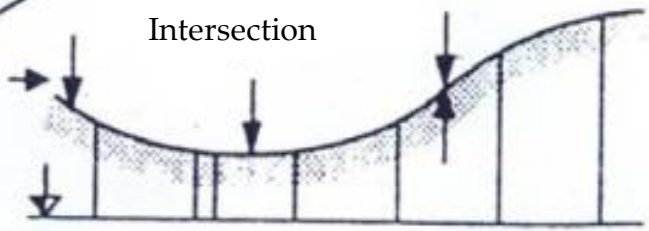


Horizantal Alignment



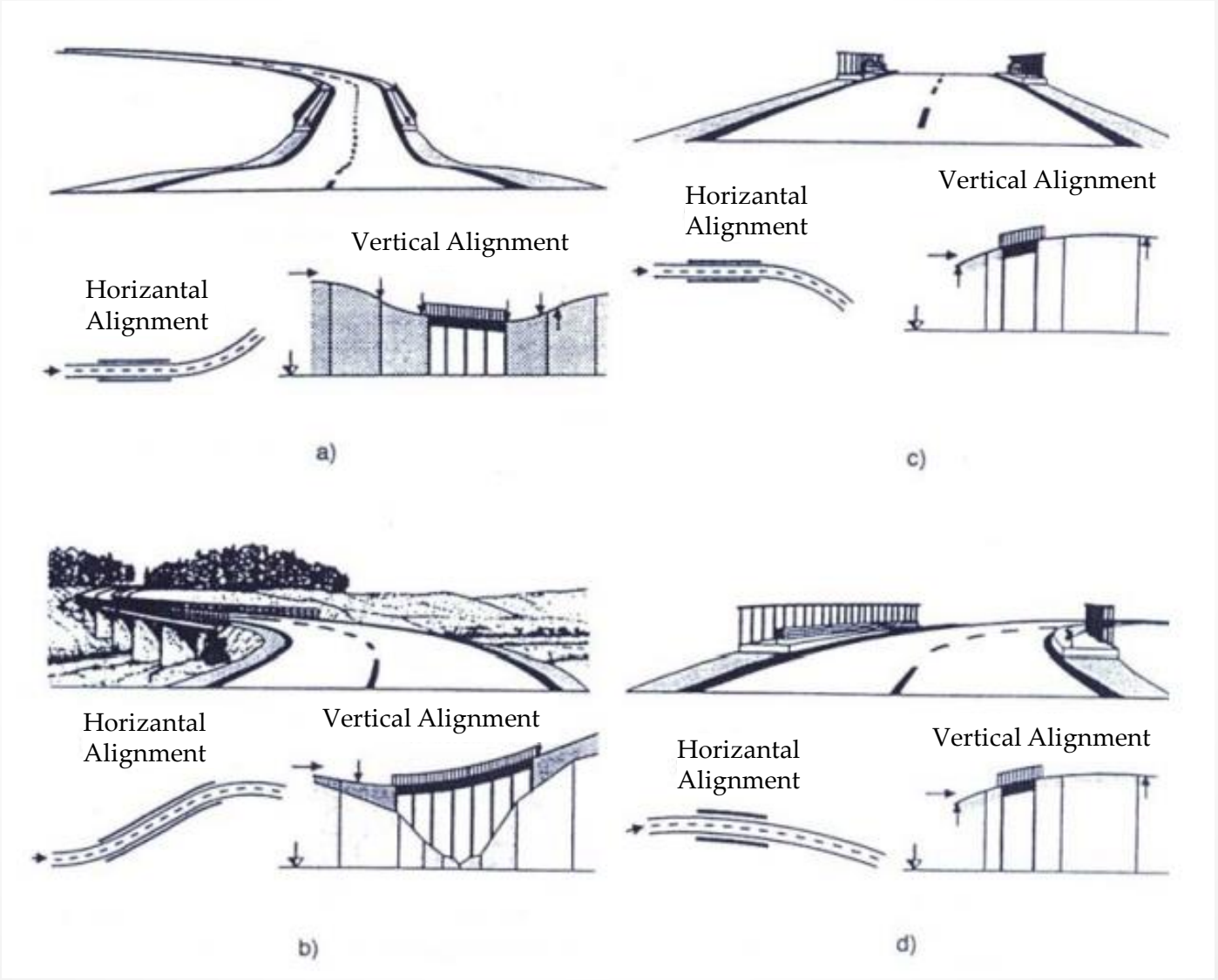
Vertical Alignment

Intersection



# Alignment and Profile Relationships in Roadway Design

Location of Bridge



# Intercestions

An intersection is defined as the general area where two or more highways join or cross, including the roadway and roadside facilities for traffic movements within the area. Each highway radiating from an intersection and forming part of it is an intersection leg. The most common intersection at which two highways cross one another has four legs. It is not recommended that an intersection have more than four legs.

Importance of Intersection;

Safety

Capacity

# Intersections

## Traffic Procedures and Conflict Points

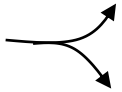
a) Diverging
b) Merging
c) Intersect
d) Weaving



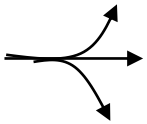
1) Right merging out (diverging)



2) Left merging out



3) Stmetrical merging out

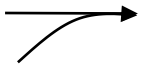


4) Multilane merging out

# Intersections

## Traffic Procedures and Conflict Points

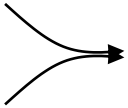
a) Diverging
b) Merging
c) Intersect
d) Weaving



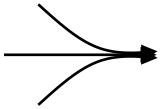
1) Rigth merging



2) Left merging



3) Symetrical merging

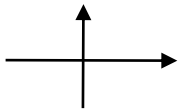


4) Multilane merging.

# Intersections

## Traffic Procedures and Conflict Points

a) Diverging
b) Merging
c) Intersect
d) Weaving

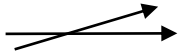


1) Perpendicalur intersect.

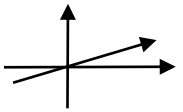
2) Angeled intercest



a) Opposite directional intersect



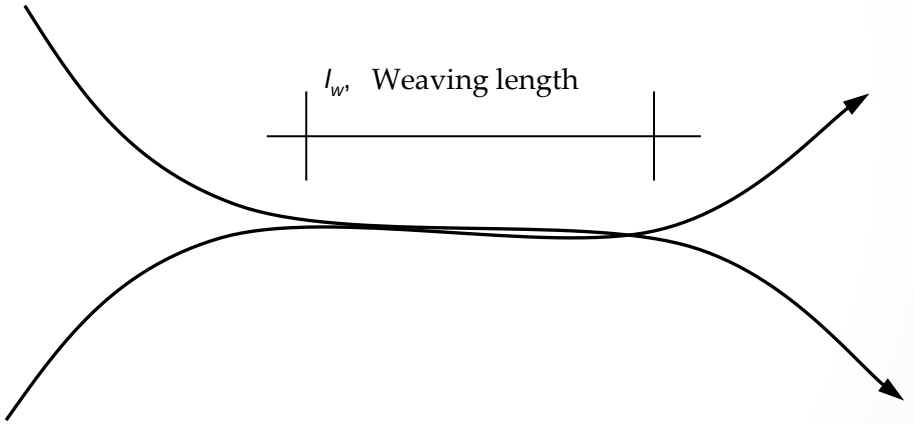
b) Same directional intersect



3) Multilane intersect

## Traffic Procedures and Conflict Points

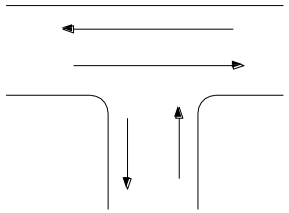
a) Diverging
b) Merging
c) Intersect
d) Weaving

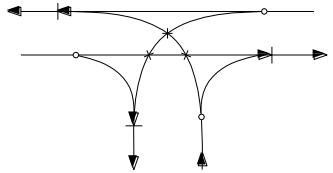


# Intersections

## Conflict points in Intersection

T Junctions

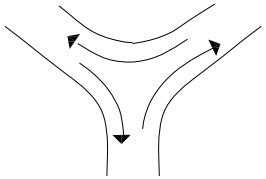


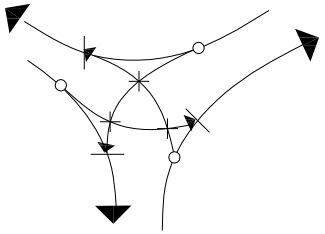


Diverging: 3
Merging: 3
Intersect: 3

Toatal =9

Y Junctions

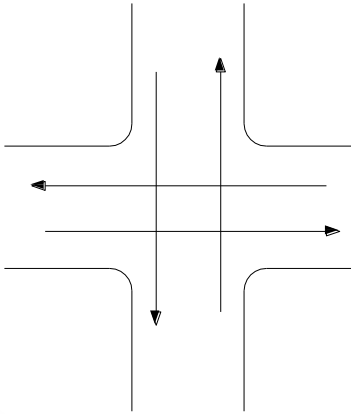


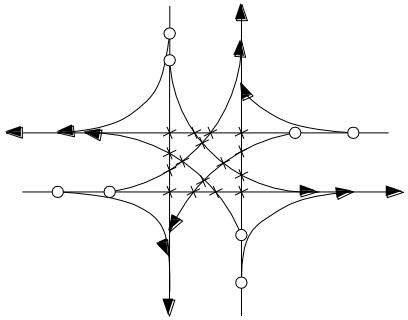


Diverging: 3
Merging: 3
Intersect: 3

TOPLAM =9

4 leg (arm) Junctions





Diverging: 8
Merging: 8
Intersect: 16

Toatal = 32

# Intercestions

## Design Consideration of Intersection

1- Increasing Safety of Intersection

2- Increasing Capacity of Intersection

3- Economic

4- Enviroment and Esthetics

5- Easy and comport of people

6- Coordinated Consequtive intersection

# Intersections

## Basic elements should be considered in intersection design

- **Human Factors**
  - Driving habits
  - Ability of drivers to make decisions
  - Driver expectancy
  - Decision and reaction time
  - Conformance to natural paths of movement
  - Pedestrian use and habits
  - Bicycle traffic use and habits

- **Traffic Considerations**
  - Design and actual capacities
  - Design-hour turning movements
  - Size and operating characteristics of vehicle
  - Variety of movements (diverging, merging, weaving, and Crossing)
  - Vehicle speeds
  - Transit involvement
  - Crash experience
  - Bicycle movements
  - Pedestrian movements

# Intersections

**Basic elements should be considered in intersection design**

- **Physical Elements**
  - Character and use of abutting property
  - Vertical alignments at the intersection
  - Sight distance
  - Angle of the intersection
  - Conflict area
  - Speed-change lanes
  - Geometric design features
  - Traffic control devices
  - Lighting equipment
  - Safety features
  - Bicycle traffic
  - Environmental factors
  - Cross walks

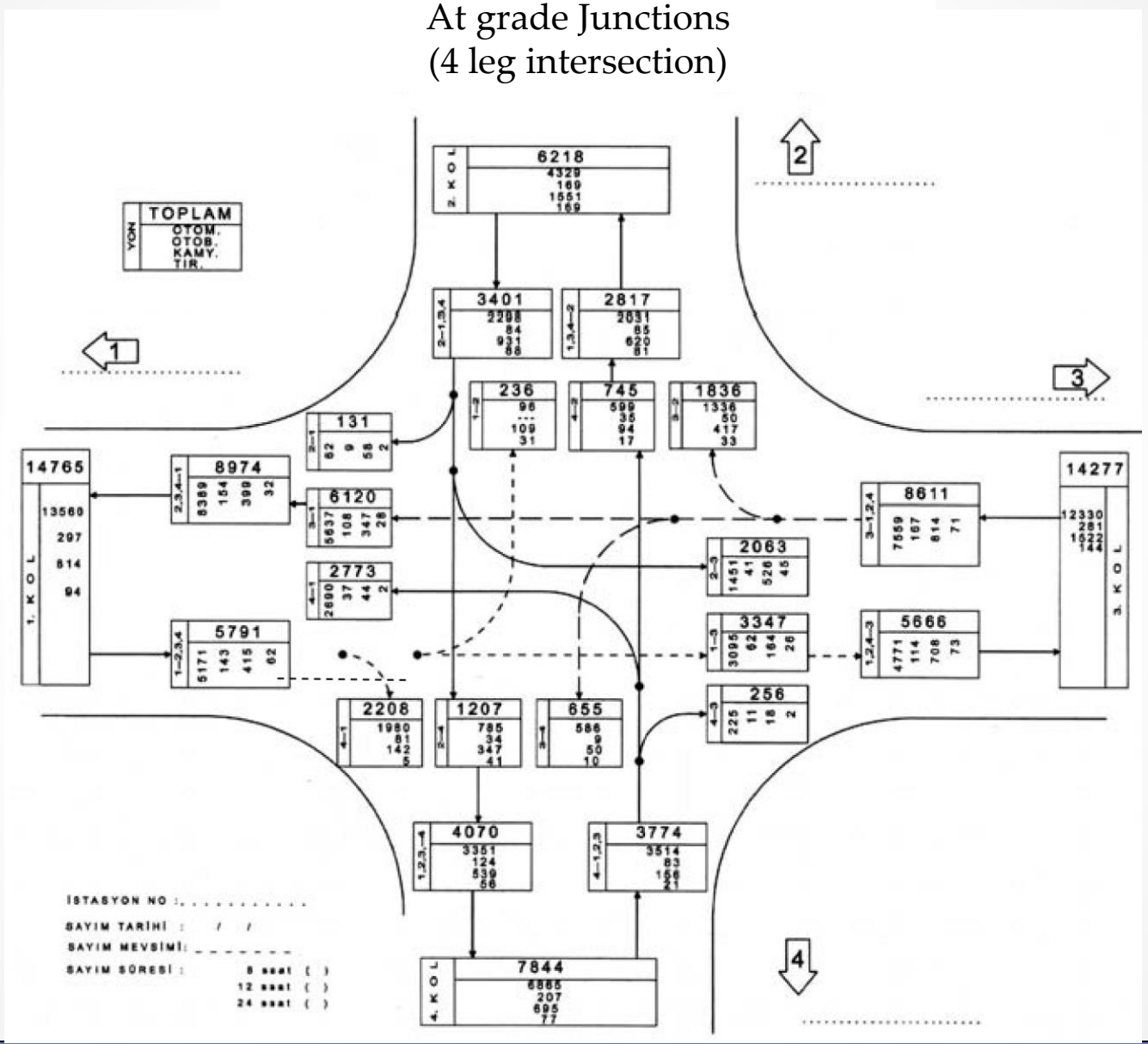
# Intersections

**Basic elements should be considered in intersection design**

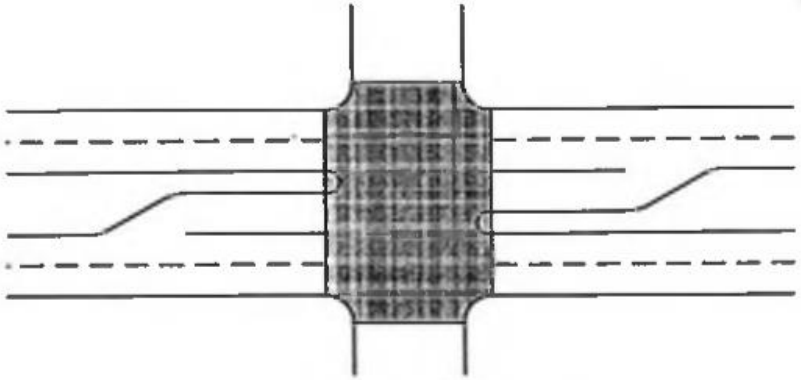
- **Economic Factors**
  - Cost of improvements
  - Effects of controlling or limiting rights-of-way on abutting residential or commercial properties where channelization restricts or prohibits vehicular movements
  - Energy consumption
- **Functional Intersection Area**

# Intersections

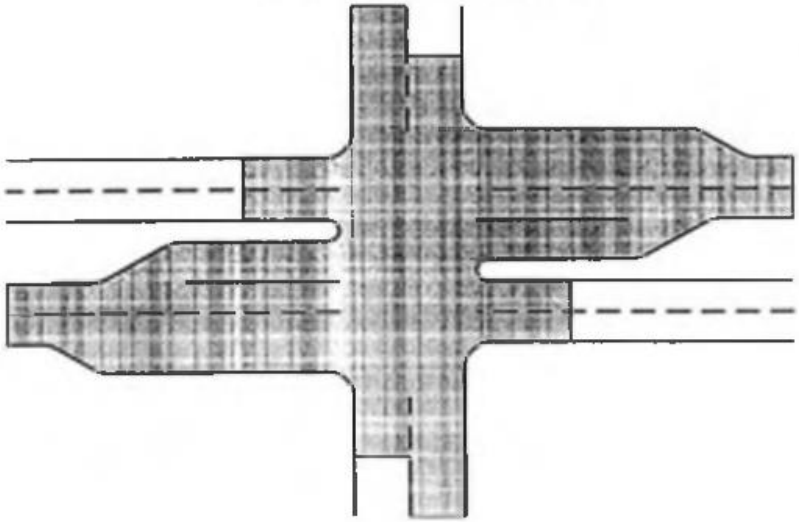
At grade Junctions
(4 leg intersection)



# Intersections



DEFINED BY PHYSICAL AREA



DEFINED BY FUNCTIONAL INTERSECTION AREA

**Exhibit 9-1. Physical and Functional Intersection Area**

# Type of Intersections

## At grade junctions

The basic types of intersections are the three-leg or T, the four-leg, and the multileg.

Clasification of at grade intersection based on:

A- Type of control:

a) Uncontrolled

b) Controlled (yield, stop)

c) Signalized

B- Type pf facility:

a) Without canalization

b) With canalization

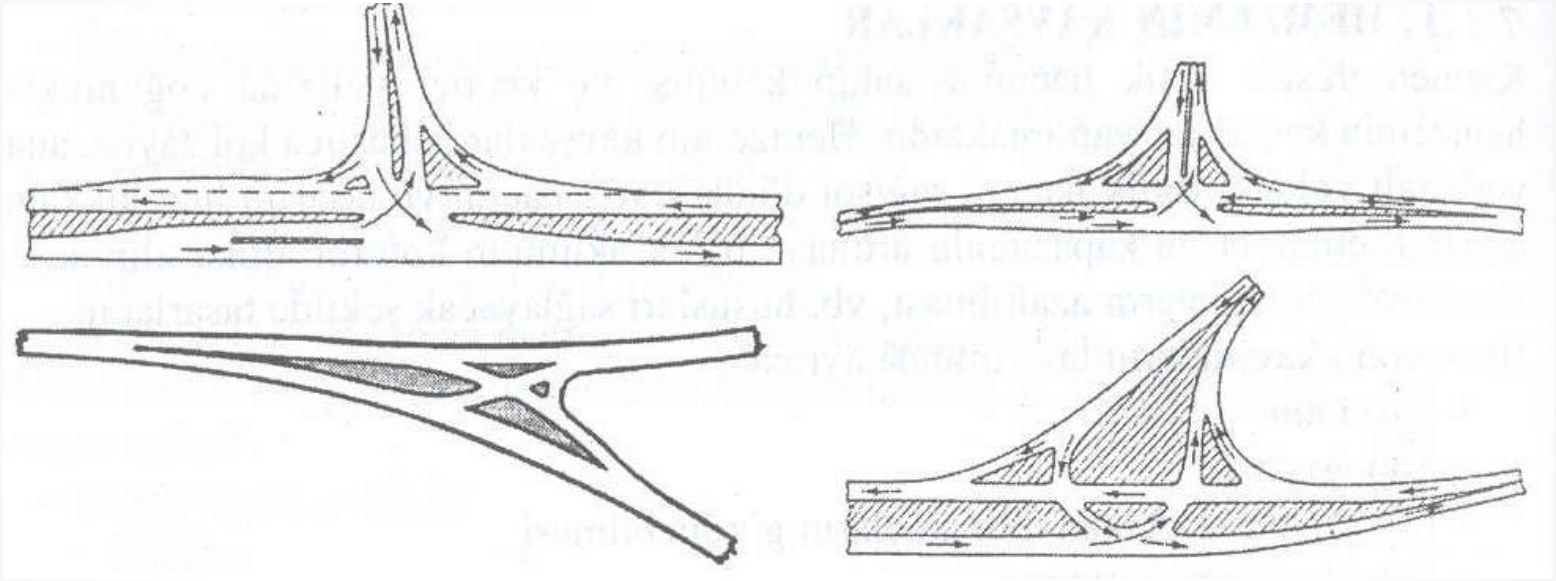
C- Type of geometry:

a) Number of legs

b) Type of leg intersection (T, Y, cross, roundabout)

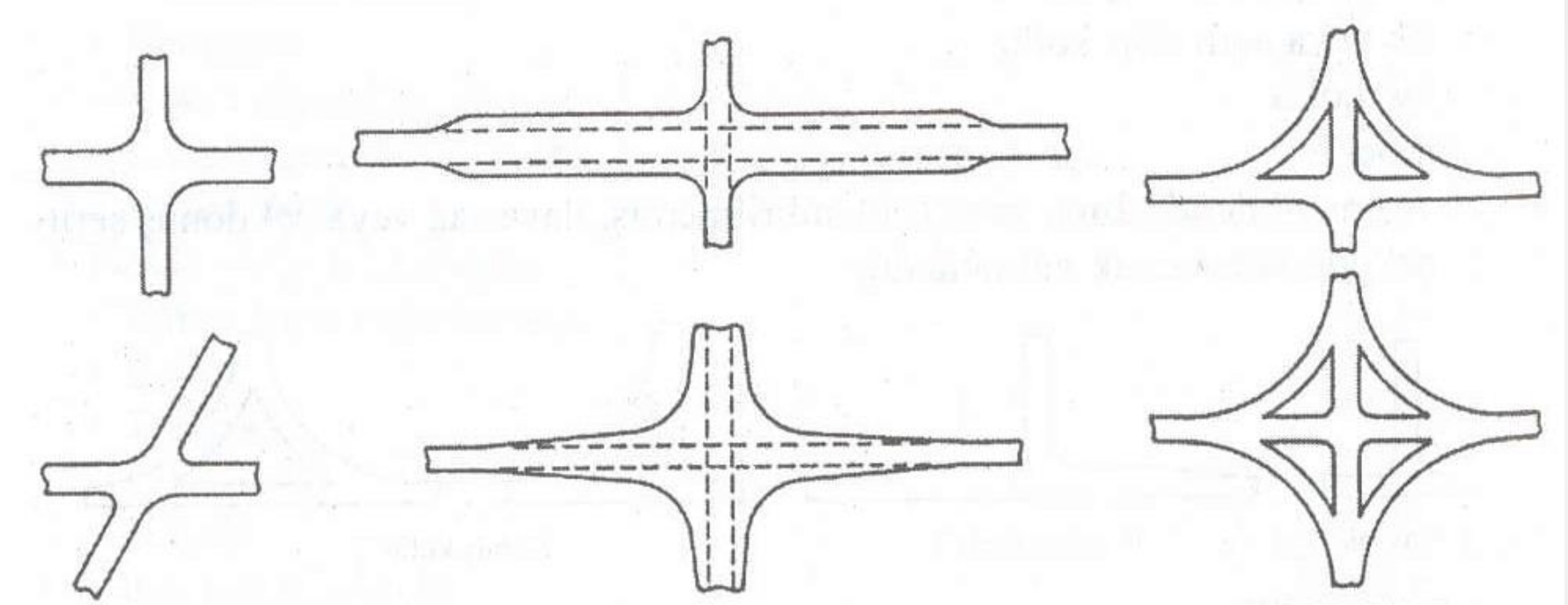
# Type of Intersections

**Canalized three leg at grade intersections**



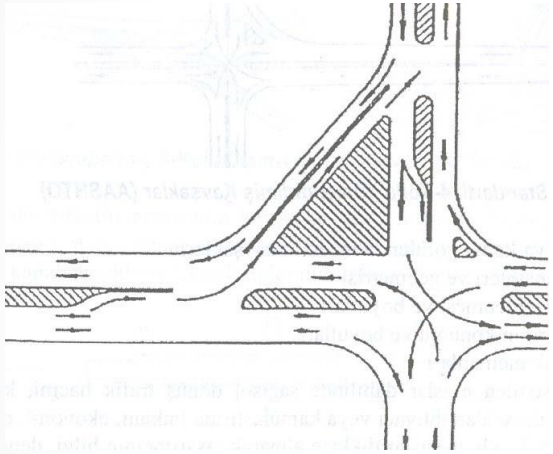
# Type of Intersections

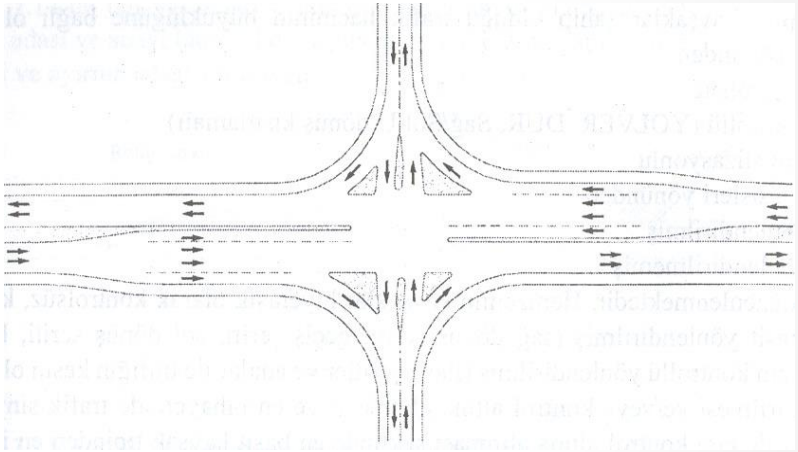
4 legs intersections



# Type of Intersections

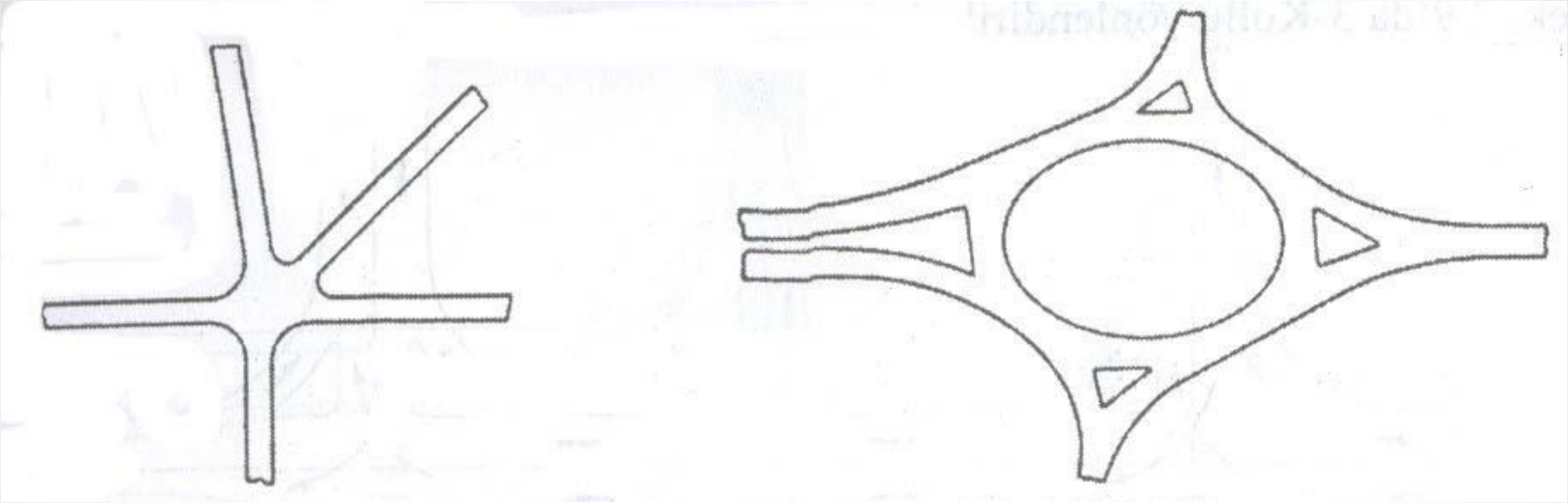
**Canalized 4 leg intersections**





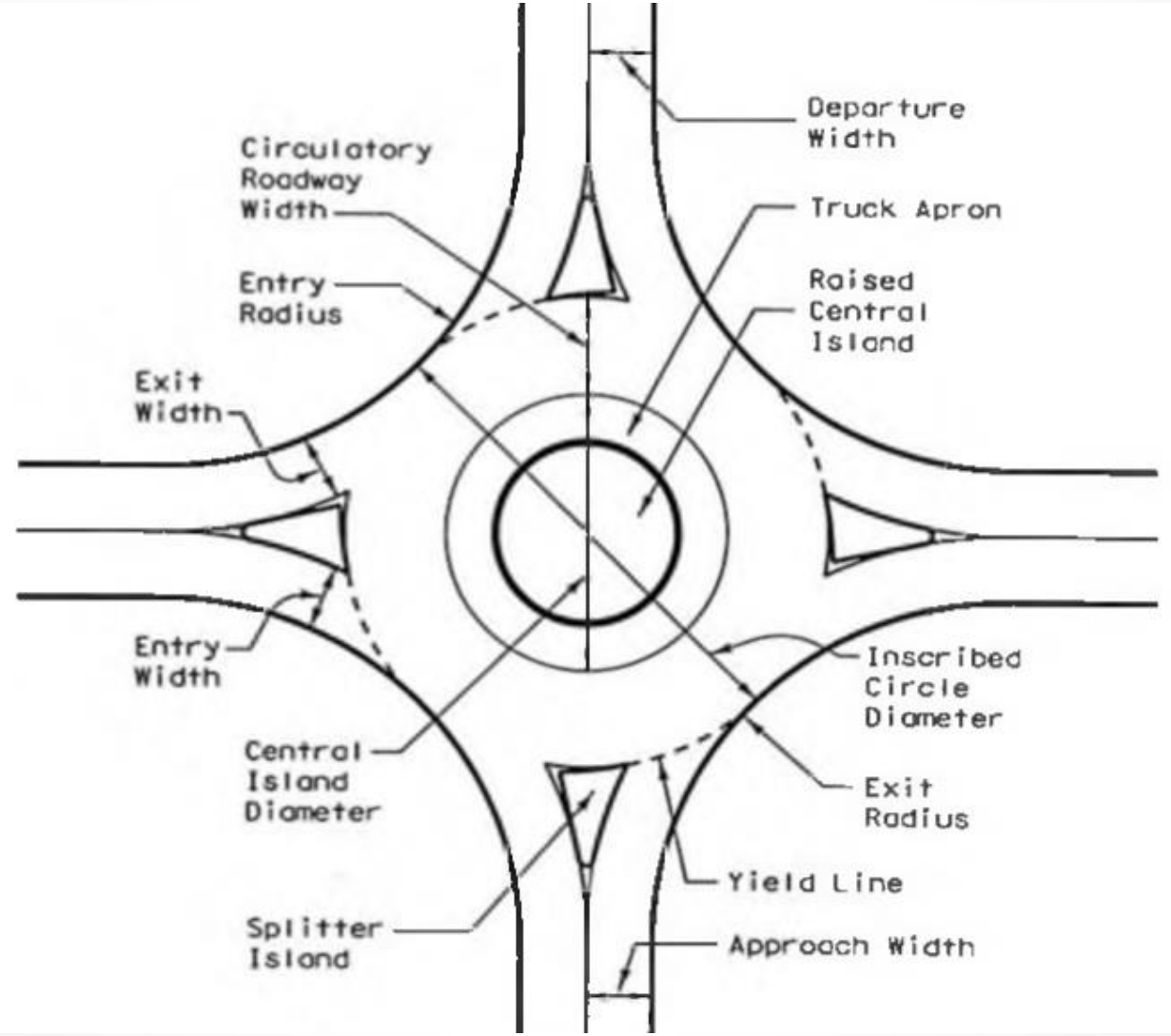
# Type of Intersections

Multileg and roundabout İntersections



# Type of Intersections

**Modern Raundabout**

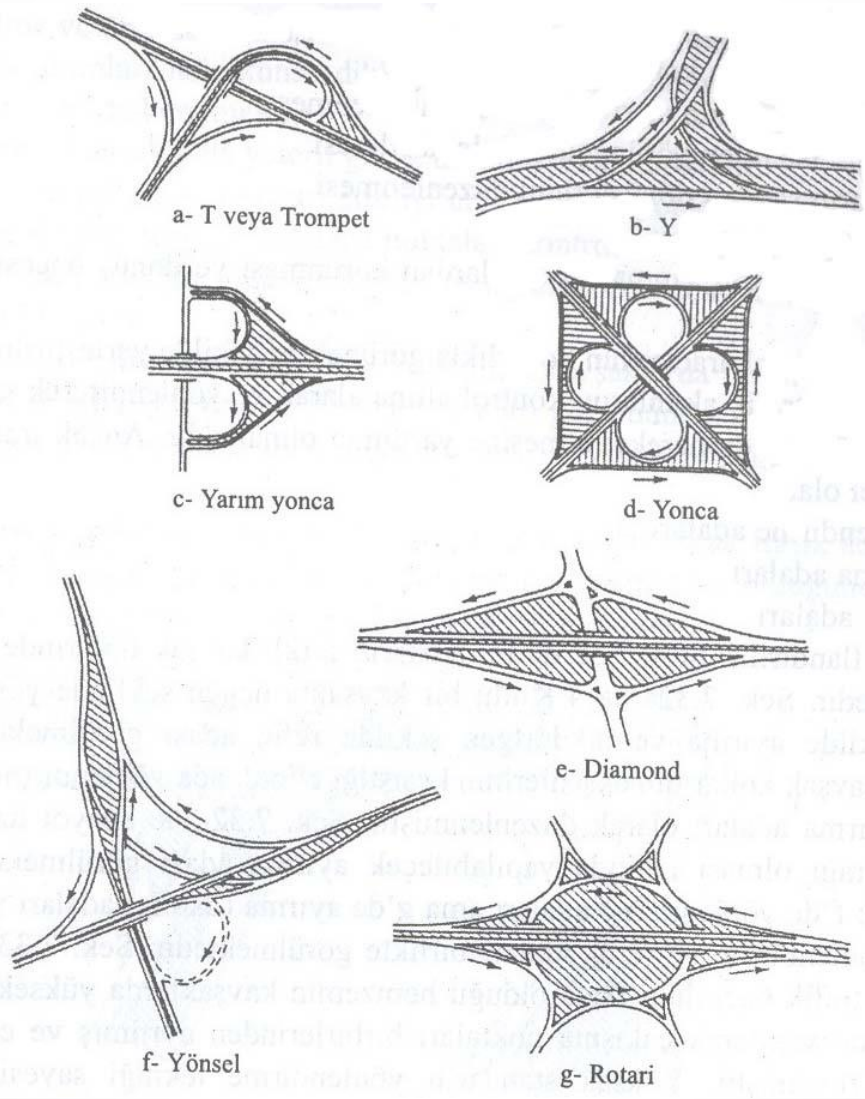


# Type of Intersections



# Type of Intersections

**Grade seperated intersections**



# Type of Intersections







# Type of Intersections

**Section 4C.01 Studies and Factors for Justifying Traffic Control Signals**

**Standard:**

01 **An engineering study of traffic conditions, pedestrian characteristics, and physical characteristics of the location shall be performed to determine whether installation of a traffic control signal is justified at a particular location.**

02 **The investigation of the need for a traffic control signal shall include an analysis of factors related to the existing operation and safety at the study location and the potential to improve these conditions, and the applicable factors contained in the following traffic signal warrants:**

**Warrant 1, Eight-Hour Vehicular Volume**
**Warrant 2, Four-Hour Vehicular Volume**
**Warrant 3, Peak Hour**
**Warrant 4, Pedestrian Volume**
**Warrant 5, School Crossing**
**Warrant 6, Coordinated Signal System**
**Warrant 7, Crash Experience**
**Warrant 8, Roadway Network**
**Warrant 9, Intersection Near a Grade Crossing**

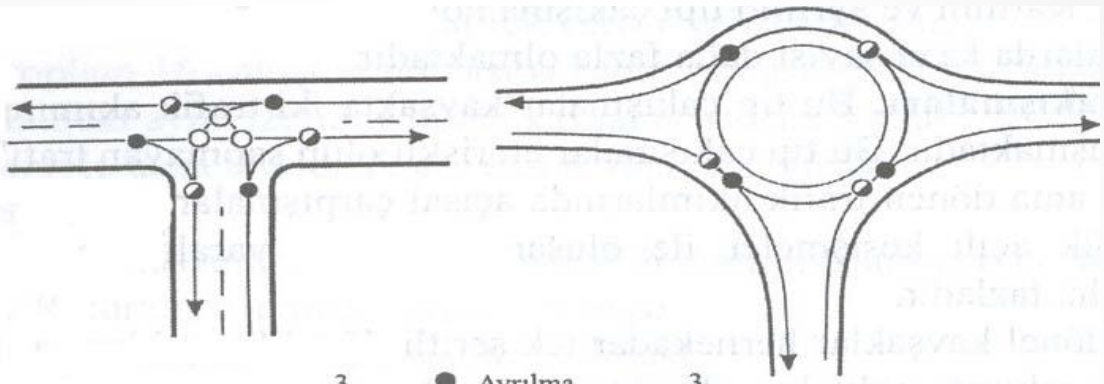
03 **The satisfaction of a traffic signal warrant or warrants shall not in itself require the installation of a traffic control signal.**

Support:

04 Sections 8C.09 and 8C.10 contain information regarding the use of traffic control signals instead of gates and/or flashing-light signals at highway-rail grade crossings and highway-light rail transit grade crossings, respectively.

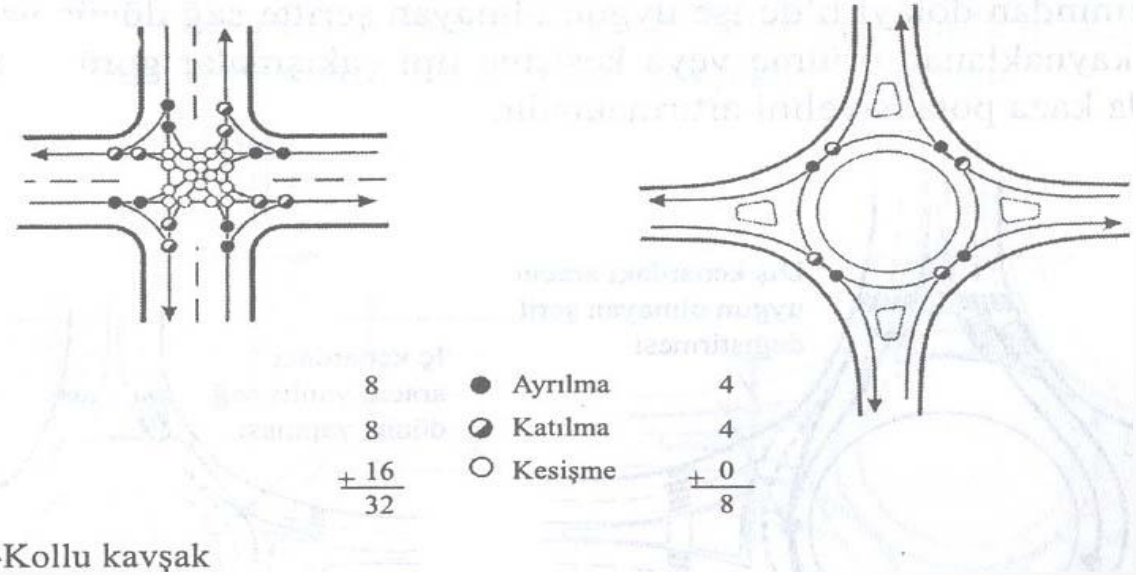
*Guidance:*

05 *A traffic control signal should not be installed unless one or more of the factors described in this Chapter are met.*



| | | |
|---|---|---|
| 3 | Ayrılma | 3 |
| 3 | Katılma | 3 |
| + 3 | Kesişme | + 0 |
| 9 | | 6 |

a- 3-Kollu kavşak



| | | |
|---|---|---|
| 8 | Ayrılma | 4 |
| 8 | Katılma | 4 |
| + 16 | Kesişme | + 0 |
| 32 | | 8 |

b- 4-Kollu kavşak